

Introduction

Great Horned Owls (*Bubo virginianus*) and Barred Owls (*Strix varia*) are both geographically widespread, nocturnal predators, with ranges that overlap primarily in the Eastern United States. Great Horned Owls are slightly larger than Barred owls but both species possess dense, soft feathers that protect them from cold and help them fly silently in pursuit of prey. However, ectoparasites like lice (order Phthiraptera), feather mites (subclass Acari), and tick flies (family Hippoboscidae) can cause sub-lethal damage to the host's feathers and feed on other tissues. Many birds and parasites have complex evolutionary traits that co-evolved due to their predator-prey relationship (Clayton 1990). It is important to monitor these parasites to assess changes in host population health.

We had two research objectives:

- Compare ectoparasite communities of each species over time by comparing historical and current parasite species richness.
- Compare parasitic communities between the two species of owls (Fig. 1).



Figure 1A. Great Horned Owl 1B. Barred Owl.

Methods

We used digitized museum specimens from the Stephen J. Taft Animal Parasitology Collection to obtain historical records of ectoparasites collected from both species of owls. We obtained owls from rehabilitation centers or as road mortality and systematically surveyed the parasites by thoroughly ruffling feathers over paper lined trays (Fig 2A). We photographed each owl and collected data on size and sex for future analyses. Paper from trays was searched on a dissecting microscope and parasites collected and identified to major taxonomic groups for comparison with museum records (Fig 2B and 2C).

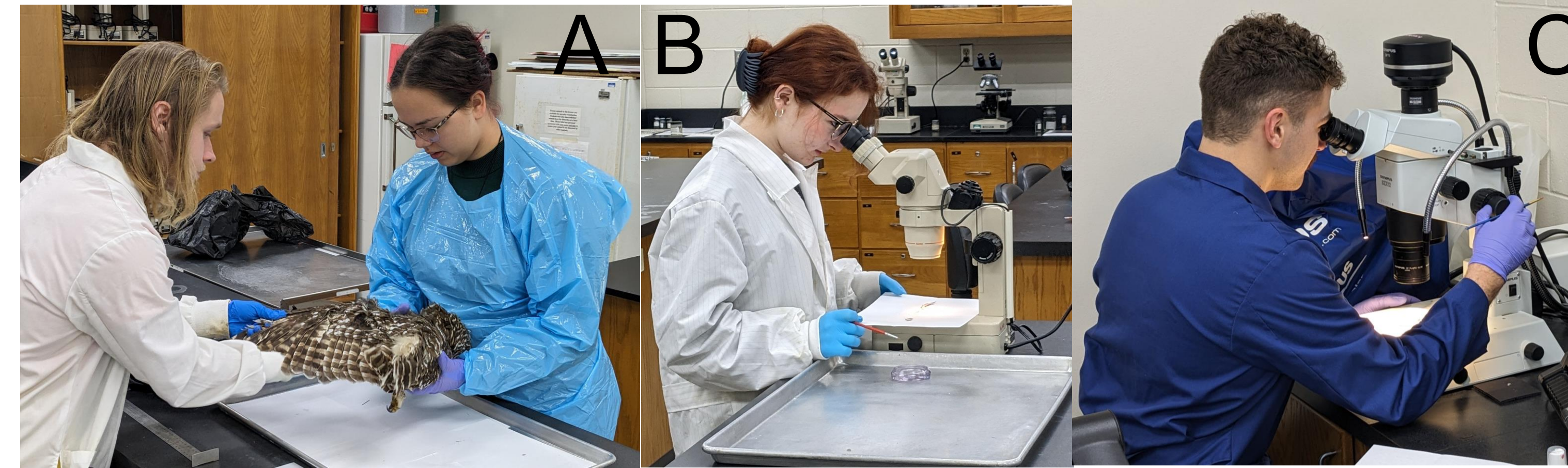


Figure 2A. Fluffing bird feathers to remove ectoparasites 2B. Checking paper for parasites 2C. Identifying parasites to taxonomic groups.

Results

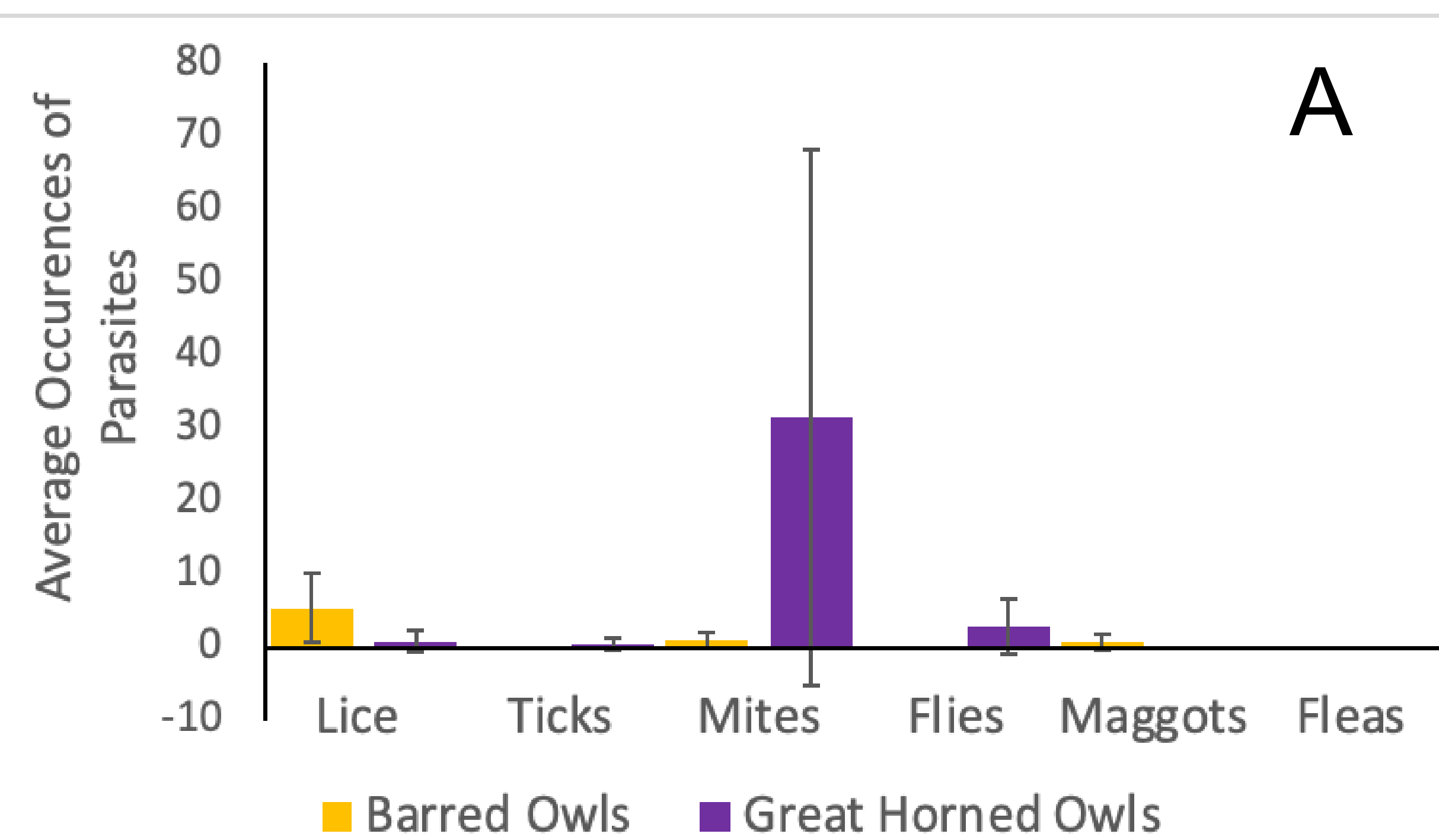
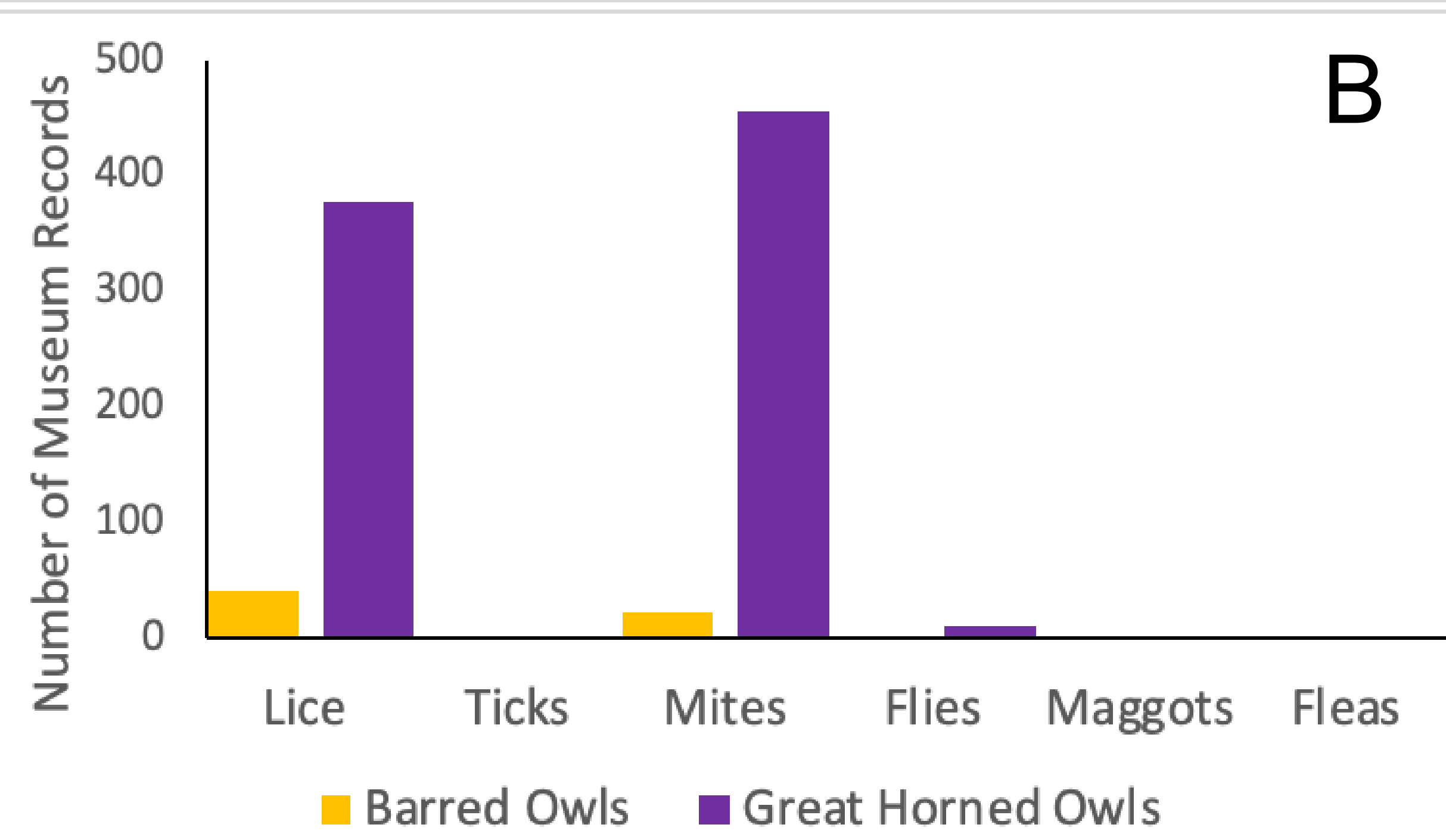


Figure 3A. Average parasite infection from new owl specimens.



3B. Number of museum records from both species of owls in the digitized collection.

Discussion

In Manitoba, CA, Great Horned Owls have higher lice densities than Barred Owls, which our data opposes (Galloway 2019). Our new results are consistent with the museum records; however, we see a decrease in the number of lice with Great Horned Owls. Our samples may underrepresent ectoparasites because these owls are donated from rehabilitation centers or as roadkill and may have lost parasites due to handling. However, the reasons for differences in parasite occurrence could be the different beak morphology between the owls. Barred Owls have recently expanded their range to the Pacific Northwest, which has resulted in hybridization with other owls (Haig et al. 2004). Tracking parasite infections can be important as owl distributions and populations change.

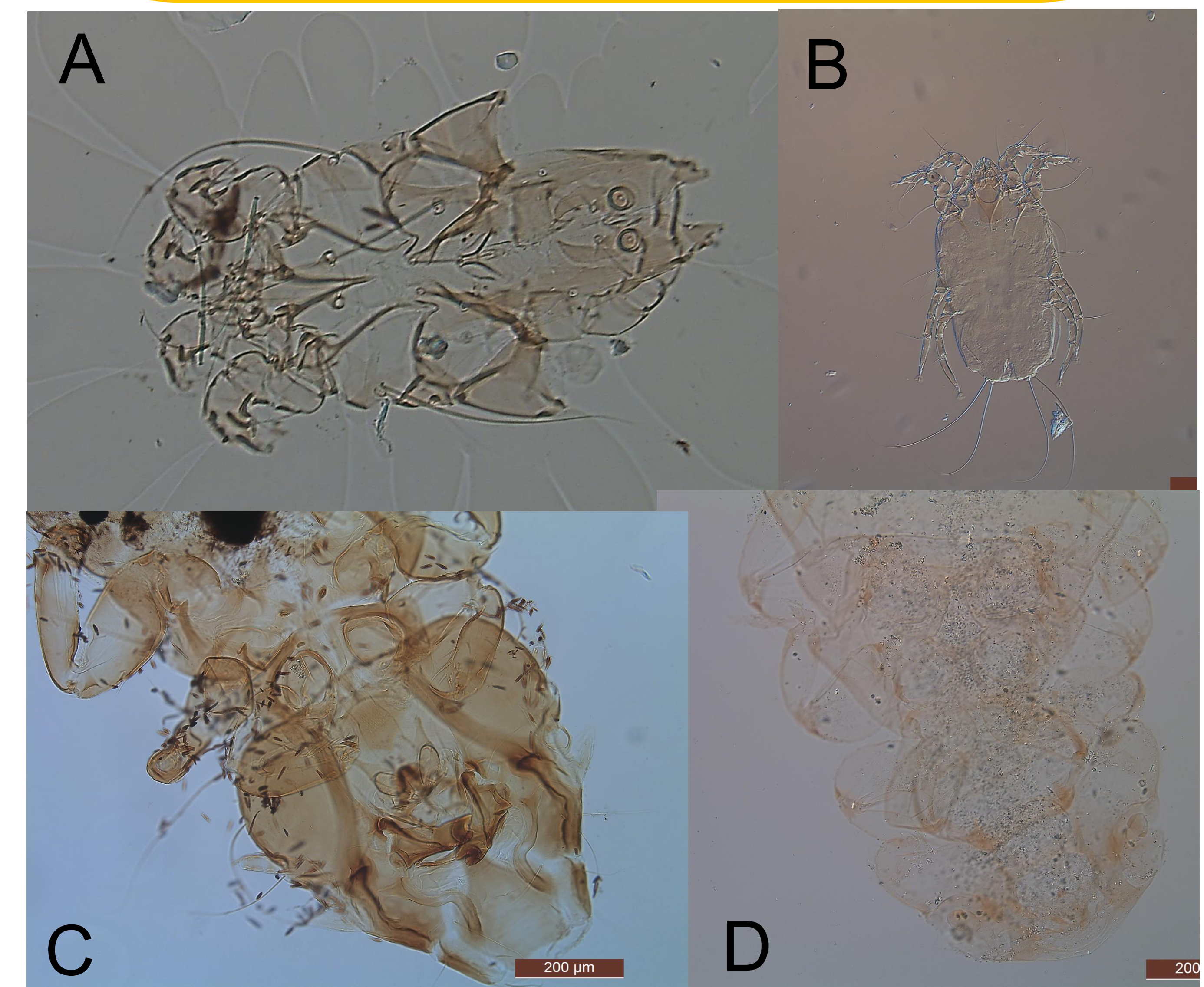


Figure 4A and 4B. Mites from museum specimens. Figure 4C. and 4D. Lice from museum specimens.

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Clayton, D. H. 1990. *Journal of Medical Entomology* 27:257-265.
Galloway, T., & Lamb, R. 2019. *The Canadian Entomologist*, 151(5), 608-620.
Haig, S.M., Mullins, T.D., Forsman, E.D., Trail, P.W. and Wennerberg, L.I.V., 2004. *Conservation Biology*, 18(5), pp.1347-1357.

There are distinct differences between the parasite communities of the two owl species based on current sampling (Fig 3A).

- Barred Owls have mostly lice and fewer mites
- Great Horned Owls have a higher average of mites than lice.

Historical museum samples revealed a similarities and differences in comparison to our current sampling (Fig 3B)

- Barred Owls are under sampled compared to Great Horned Owls
- Barred Owls had lice and mites but also one record of a flea
- Great Horned Owls had both large numbers of mites and lice